CPE301 – SPRING 2024

Design Assignment 6

Student Name: Joshua Martinez
Student #: 5007531628
Student Email: marti87@unlv.nevada.edu
Primary Github address: https://github.com/JoshuaMa2003
Directory: https://github.com/JoshuaMa2003/submission_da
Video Playlist:
https://www.youtube.com/playlist?list=PLZ09MA_uFt63me60r5XWtDUj2m21n4nUP

PLEASE NOTE: AS OF SUBMITTING THIS ASSIGNMENT ON 5/4/2024, MY ATMEGA328PB MICROCONTROLLER WAS NOT FUNCTIONING, SO I WAS NOT ABLE TO GET ANY VIDEOS OR PICTURES OF MY BOARD

1. COMPONENTS LIST AND DIAGRAM SHOWING PINS USED

List of Components used

- **Microchip Studio Debugger**
- **Microchip Studio Simulator**
- **ATmega328PB Microcontroller**
- **DC Motor**
- **ROB-14450 Motor Driver**
- **ATmega328PB Microcontroller Potentiometer**
- **Male-to-Male Wires**
- **Multifunctional Shield Seven Segment Display**

2. INITIAL CODE OF TASK

```
#define F_CPU 16000000UL          //define the clock frequency as 16 mhz
#include <avr/io.h>               //include the input/output library for avr
#define ENABLE 3                  //define enable pin as pb3
#define MTR_1 1                   //define motor 1 pin as pb1
#define MTR_2 2                   //define motor 2 pin as pb2
#define SW (PIND & (1<<7))        //read switch state from pd7
#include <util/delay.h>           //include the delay functions
#include <stdio.h>                //include the standard input/output library
#include <avr/interrupt.h>        //include the interrupt library

//function declarations
void read_adc(void);
void adc_init(void);

volatile unsigned int adc_temp;     //variable to store adc result

//initialize adc settings
void adc_init(void)
{
        ADMUX = (0<<REFS1)|           //use avcc as reference voltage
        (1<<REFS0)|
        (0<<ADLAR)|           //right adjust adc result
        (0<<MUX2)|            //use adc0 (pc0, pin23)
        (0<<MUX1)|
        (0<<MUX0);
        ADCSRA = (1<<ADEN)|           //enable adc
        (0<<ADSC)|            //do not start conversion
        (0<<ADATE)|           //disable auto trigger
        (0<<ADIF)|            //clear adc interrupt flag
        (0<<ADIE)|            //disable adc interrupt
        (1<<ADPS2)|           //set adc prescaler to 128
        (0<<ADPS1)|
        (1<<ADPS0);
}

//read adc value
void read_adc(void)
{
        unsigned char i = 4;
        adc_temp = 0;
        while (i--)
        {
```

```
            ADCSRA |= (1<<ADSC);        //start conversion
            while (ADCSRA & (1<<ADSC));  //wait for conversion to finish
            adc_temp += ADC;           //sum up results
            _delay_ms(50);             //wait 50 ms
        }
        adc_temp = adc_temp / 4;        //average four samples
        adc_temp = adc_temp * 50 / 1024;  //convert to voltage scale
}

int main(void)
{
        adc_init();                 //initialize adc
        PORTD |= (1<<7);            //enable pull-up resistor on pd7
        DDRB |= 0b00001110;         //set pb3, pb1, pb2 as outputs
        PORTB &= ~(1<<ENABLE);      //disable motor initially
        PORTB &= ~(1<<MTR_1);       //turn off mtr_1
        PORTB &= ~(1<<MTR_2);       //turn off mtr_2
        DDRB |= (1<<3);             //set oc2a as output
        OCR2A = 50;                 //set output compare for pwm

        //configure timer 2 for fast pwm
        TCCR2A = (1<<COM2A1)|(1<<WGM21)|(1<<WGM20);
        TCCR2B = 0x02;              //set prescaler to 8

        while (1)
        {
            read_adc();             //read adc value
            PORTD |= (1<<ENABLE);       //enable motor
            if(SW != 0)             //if switch is on
            {
                _delay_ms(20);          //delay for 20ms
                PORTB |= (1<<MTR_1);    //drive mtr_1 high for clockwise rotation
                PORTB &= ~(1<<MTR_2);   //keep mtr_2 low
            }
            else
            {
                _delay_ms(20);          //delay for 20ms
                PORTB &= ~(1<<MTR_1);   //drive mtr_1 low
                PORTB |= (1<<MTR_2);    //drive mtr_2 high for counter-clockwise
rotation
            }
            OCR2A = adc_temp * 15;      //adjust pwm based on adc result
        }
}
```

2. INITIAL CODE OF TASK

```
/*
 * CPE301_DesignAssignment6_Task2.c
 *
 * Created: 5/4/2024 10:46:33 PM
 * Author : joshu
 */

#define F_CPU 16000000UL          //define the clock frequency as 16 mhz
#include <avr/io.h>               //include the input/output library for avr
#include <util/delay.h>           //include the delay functions
#include <stdio.h>                //include the standard input/output library
#include <avr/interrupt.h>        //include the interrupt library

#define ENABLE 3                  //define enable pin as pb3
#define MTR_1 1                   //define motor 1 pin as pb1
#define MTR_2 2                   //define motor 2 pin as pb2
#define SW (PIND & (1<<7))        //read switch state from pd7

#define BAUDRATE 9600             //define default baud rate for serial communication
#define BAUD_PRESCALLER (((F_CPU / (BAUDRATE * 16UL))) - 1)  //calculate prescaler for uart

//function declarations
void read_adc(void);
void adc_init(void);
void USART_init(unsigned int ubrr);
void USART_tx_string(char *data);
void InitTimer1(void);
void InitExtInter(void);
void StartTimer1(void);

//global variables for adc, timer counts, etc.
volatile unsigned int adc_temp, revTickAvg, revCtr, ticks, timeTicks, timeCount;
volatile uint32_t tickv, T1Ovs2;
char outs[20];
char please[5] = "1.1\n\0";

//timer 0 overflow interrupt service routine
ISR(TIMER0_OVF_vect) {
        timeTicks++;
        if (timeTicks > 980) {
                snprintf(outs, sizeof(outs), "%.2d\n", ticks / 24);  //format ticks to string
                please[0] = outs[0];
                please[2] = outs[1];
```

```
                USART_tx_string("RPS: ");    //send rpm data via uart
                USART_tx_string(please);
                ticks = 0;
                timeCount = 0;
                timeTicks = 0;
        }
        TCNT0 = 0;  //reset timer counter
}

//external interrupt 1 service routine for counting motor revolutions
ISR(INT1_vect) {
        ticks++;
}

//adc initialization
void adc_init(void) {
        ADMUX = (0<<REFS1) | (1<<REFS0) | (0<<ADLAR) | (0<<MUX2) | (0<<MUX1) |
(0<<MUX0);
        ADCSRA = (1<<ADEN) | (0<<ADSC) | (0<<ADATE) | (0<<ADIF) | (0<<ADIE) | (1<<ADPS2)
| (0<<ADPS1) | (1<<ADPS0);
}

//read adc value
void read_adc(void) {
        unsigned char i = 4;
        adc_temp = 0;
        while (i--) {
                ADCSRA |= (1<<ADSC);
                while (ADCSRA & (1<<ADSC));
                adc_temp += ADC;
                _delay_ms(50);
        }
        adc_temp = adc_temp / 4;  //average four samples
        adc_temp = adc_temp * 50 / 1024; //convert to voltage
}

//timer 1 initialization
void InitTimer1(void) {
        TCCR0A = 0;
        TCNT0 = 255;
        TIMSK0 = (1 << TOIE0);  //enable timer0 overflow interrupt
        TCCR0B |= (1 << CS01) | (1 << CS00);  //set prescaler to 64
        sei();  //enable global interrupts
}
```

```
//external interrupt setup
void InitExtInter(void) {
	DDRD &= ~(1 << 2);  //set pd2 as input
	PORTD |= (1 << 1) | (1 << 2);  //enable pull-ups
	EICRA = (1 << 2);  //set int1 to trigger on falling edge
	EIMSK = (1 << INT1);  //enable int1
	sei();  //enable global interrupts
}

//usart initialization
void USART_init(unsigned int ubrr) {
	UBRR0H = (unsigned char)(ubrr >> 8);
	UBRR0L = (unsigned char)ubrr;
	UCSR0B = (1 << TXEN0);  //enable transmitter
	UCSR0C = (3 << UCSZ00);  //set frame format: 8 data bits, no parity, 1 stop bit
}

//send string over usart
void USART_tx_string(char *data) {
	while (*data != '\0') {
		while (!(UCSR0A & (1 << UDRE0)));  //wait for empty transmit buffer
		UDR0 = *data++;  //put data into buffer, sends the data
	}
}

int main(void) {
	InitExtInter();
	InitTimer1();
	adc_init();			//initialize adc
	USART_init(BAUD_PRESCALLER);	//initialize usart
	USART_tx_string("Connected!\r\n");  //send connection message
	_delay_ms(125);			//wait a bit

	PORTD |= (1 << 7);		//enable pull-up on pd7
	DDRB |= 0b00001110;		//set pb3, pb1, pb2 as outputs
	PORTB &= ~(1 << ENABLE);	//disable motor initially
	PORTB &= ~(1 << MTR_1);		//motor 1 off
	PORTB &= ~(1 << MTR_2);		//motor 2 off
	DDRB |= (1 << 3);		//set oc2a as output
	OCR2A = 50;			//set output compare for pwm
	TCCR2A = (1 << COM2A1) | (1 << WGM21) | (1 << WGM20);  //configure for fast pwm
	TCCR2B = 0x02;			//set prescaler to 8
```

```
        while (1) {
                read_adc();  //read adc value continuously
                PORTD |= (1 << ENABLE);  //enable motor
                if (SW != 0) {  //if switch is on
                        _delay_ms(20);
                        PORTB |= (1 << MTR_1);  //motor 1 on for clockwise rotation
                        PORTB &= ~(1 << MTR_2);  //motor 2 off
                        } else {
                        _delay_ms(20);
                        PORTB &= ~(1 << MTR_1);  //motor 1 off
                        PORTB |= (1 << MTR_2);  //motor 2 on for counter-clockwise rotation
                }
                OCR2A = adc_temp * 15;  //adjust pwm based on adc result
        }
}
```

2. INITIAL CODE OF TASK

```
/*
 * CPE301_DesignAssignment6_Task3.c
 *
 * Created: 5/4/2024 10:53:31 PM
 * Author : joshu
 */

#define F_CPU 16000000UL          //define clock frequency as 16 MHz for delay calculations
#include <avr/io.h>               //include AVR device-specific IO definitions
#include <util/delay.h>           //include functions for delay loops
#include <stdio.h>                //include standard input and output functions
#include <avr/interrupt.h>        //include interrupt handling functionality

#define ENABLE 3                  //define enable pin number for motor control
#define MTR_1 1                   //define motor 1 control pin
#define MTR_2 2                   //define motor 2 control pin
#define SW (PIND&(1<<7))          //define switch input using pin PD7
#define SHIFT_REGISTER DDRB       //define shift register direction register
#define SHIFT_PORT PORTB          //define shift register port
#define DATA (1<<PB3)             //define data input pin for SPI (MOSI)
#define LATCH (1<<PB2)            //define latch pin for SPI (SS)
#define CLOCK (1<<PB5)            //define clock pin for SPI (SCK)

#define BAUDRATE 9600             //define baud rate for USART
#define BAUD_PRESCALLER (((F_CPU / (BAUDRATE * 16UL))) - 1)  //calculate USART baud rate
prescaler

//function prototypes
void read_adc(void);
void adc_init(void);
void USART_init(unsigned int ubrr);
void USART_tx_string(char *data);
void init_IO(void);
void init_SPI(void);
void spi_send(unsigned char byte);

//global variables
volatile unsigned int adc_temp, revTickAvg, ticks, recTicks, recTicks2, timeTicks, timeCount;
char outs[20];
char please[5] = "1.1\n\0";
const uint8_t SEGMENT_MAP[] = {0xC0, 0xF9, 0xA4, 0xB0, 0x99, 0x92, 0x82, 0xF8, 0x80,
0x90}; //segment byte maps for numbers 0-9
const uint8_t SEGMENT_SELECT[] = {0xF1, 0xF2, 0xF4, 0xF8}; //byte maps to select digit 1-4
```

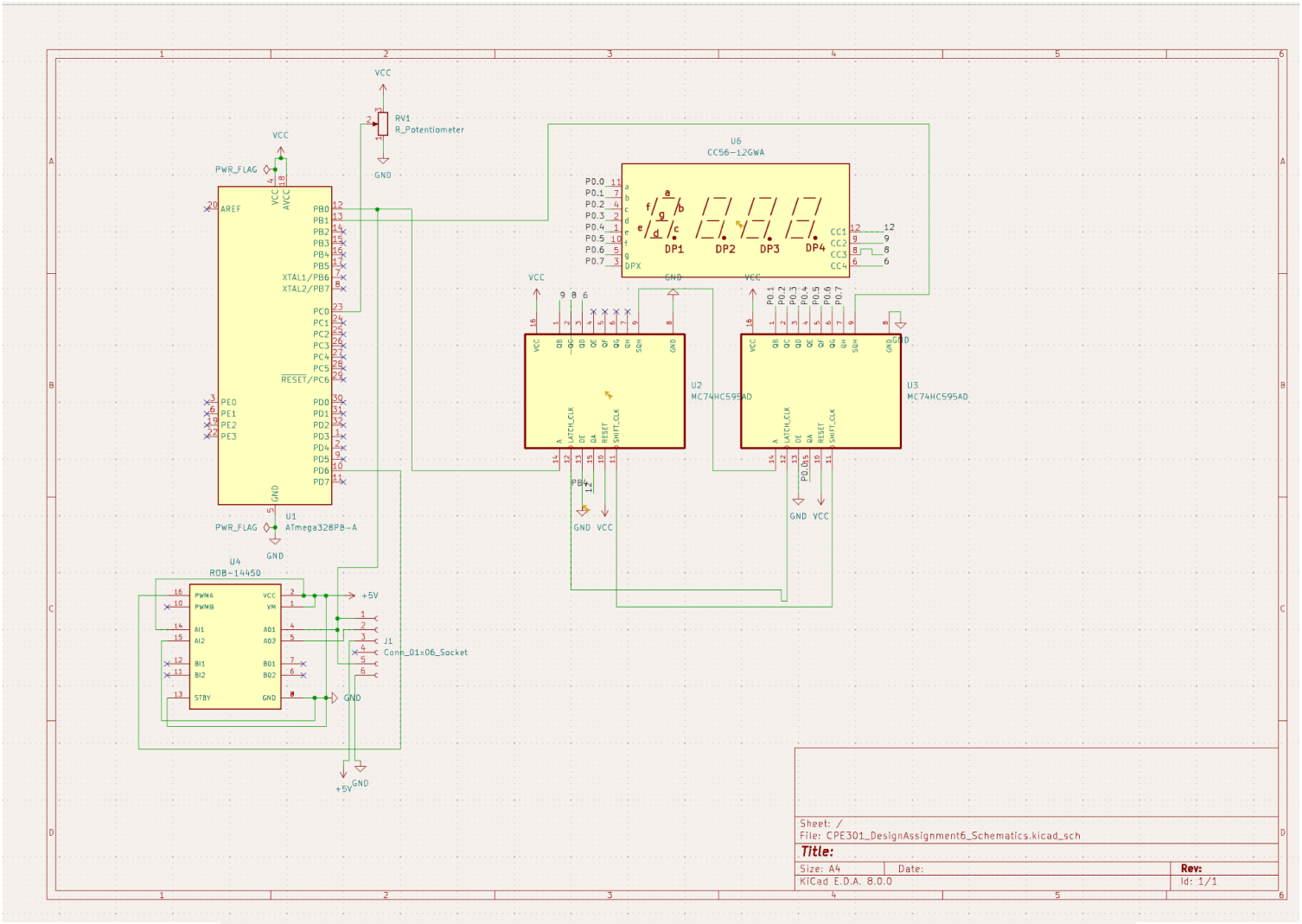
```
volatile int alternator = 0;

//setup IO for SPI and shift register
void init_IO(void){
	SHIFT_REGISTER |= (DATA | LATCH | CLOCK); //set control pins as outputs
	SHIFT_PORT &= ~(DATA | LATCH | CLOCK); //initialize control pins to low
}

//initialize SPI in Master mode
void init_SPI(void){
	SPCR0 = (1<<SPE) | (1<<MSTR); //enable SPI and set as Master
}

//send byte via SPI
void spi_send(unsigned char byte){
	SPDR0 = byte; //load byte to SPI data register
	while(!(SPSR0 & (1<<SPIF))); //wait until SPI transmission is complete
}

//ISR for TIMER0 overflow
ISR(TIMER0_OVF_vect) {
	timeTicks++;
	if(timeTicks > 980){
		recTicks = ticks;
		ticks = 0;
		timeCount = 0;
		timeTicks = 0;
	}
	TCNT0 = 0;
}

//ISR for external interrupt 0
ISR(INT0_vect){
	ticks++;
}

//initialize ADC
void adc_init(void){
	ADMUX = (0<<REFS1)|(1<<REFS0)|(0<<ADLAR)|(0<<MUX2)|(0<<MUX1)|(0<<MUX0);
//select AVcc as reference, right adjust, select ADC0
	ADCSRA =
(1<<ADEN)|(0<<ADSC)|(0<<ADATE)|(0<<ADIF)|(0<<ADIE)|(1<<ADPS2)|(0<<ADPS1)|(1<<ADP
S0); //enable ADC, set prescaler to 128
}
```

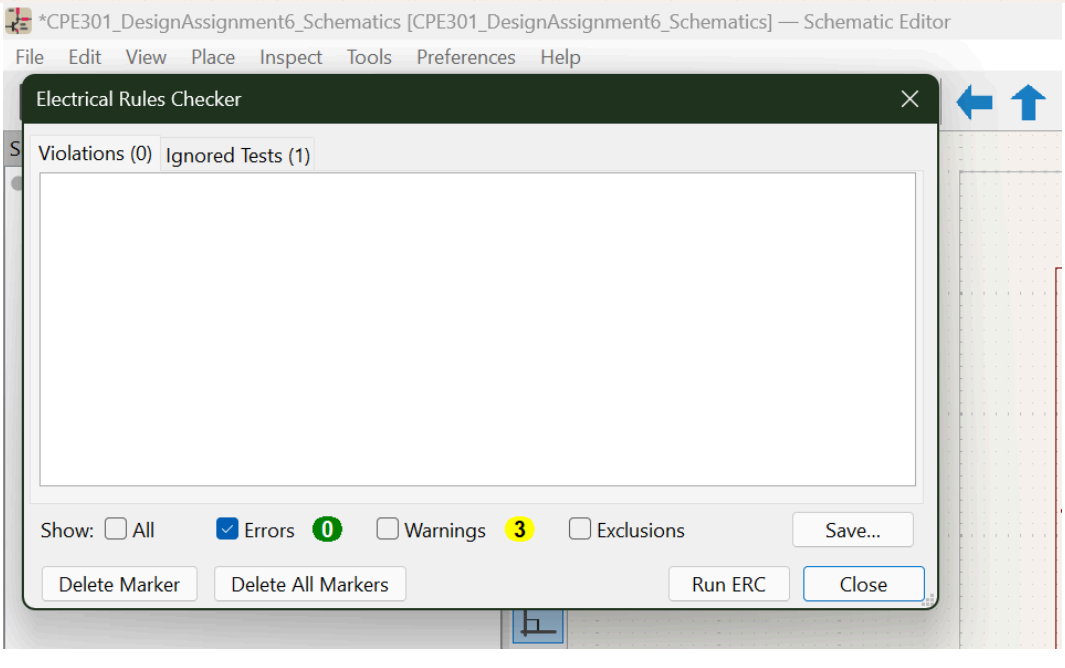
```
//read ADC value
void read_adc(void){
	unsigned char i = 4;
	adc_temp = 0;
	while (i--){
		ADCSRA |= (1<<ADSC); //start ADC conversion
		while(ADCSRA & (1<<ADSC)); //wait for conversion to complete
		adc_temp += ADC; //accumulate converted result
		_delay_ms(50);
	}
	adc_temp = adc_temp / 4; //average over four samples
	adc_temp = adc_temp * 50 / 1024; //convert to voltage
}

//initialize timer0 for periodic interrupts
void InitTimer1(void) {
	TCCR0A = 0;
	TCNT0 = 255; //preload timer counter
	TIMSK0 = (1 << TOIE0); //enable overflow interrupt
	TCCR0B |= (1 << CS01) | (1 << CS00); //set prescaler to 64
	sei(); //enable global interrupts
}

//setup external interrupts
void InitExtInter(void){
	DDRD &= ~(1 << 2); //set PD2 as input
	PORTD |= (1 << 1) | (1<<2); //enable pull-up resistors
	EICRA = (1<<2); //trigger INT0 on falling edge
	EIMSK = (1<<INT0); //enable external interrupt 0
	sei(); //enable global interrupts
}

//initialize USART for serial communication
void USART_init(unsigned int ubrr){
	UBRR0H = (unsigned char)(ubrr>>8);
	UBRR0L = (unsigned char)ubrr;
	UCSR0B = (1 << TXEN0); //enable transmitter
	UCSR0C = (3 << UCSZ00); //set frame format: 8 data bits, no parity, 1 stop bit
}

//transmit string via USART
void USART_tx_string(char *data){
	while ((*data != '\0')){
```

```
            while (!(UCSR0A & (1 <<UDRE0))); //wait for empty transmit buffer
            UDR0 = *data; //send character
            data++;
        }
}

int main(void){
        InitExtInter();
        InitTimer1();
        adc_init(); //initialize ADC
        init_IO();
        init_SPI();
        DDRB |= 0b00001110; //set PB3, PB1, PB2 as outputs for motor control
        DDRD |= (1<<ENABLE) | (1<<5) | (1<<6); //set pins as outputs
        OCR2A = 50; //set PWM duty cycle
        TCCR2A = (1<<COM2B1)|(1<<WGM21)|(1<<WGM20); //configure Timer2 for fast
PWM
        TCCR2B = 0x02; //set prescaler to 8

        while(1){
            read_adc();
            PORTD |= (1<<ENABLE); //enable motor
            PORTD |= (1<<5); //set MTR_1 high
            PORTD &= ~(1<<6); //set MTR_2 low

            OCR2B = adc_temp * 15; //adjust PWM based on ADC reading

            // Display handling in the SPI block
            if(alternator == 1){
                recTicks2 = recTicks;
                SHIFT_PORT &= ~LATCH; //pull LATCH low to start SPI transfer
                if(recTicks2/24 >= 10){
                    spi_send((unsigned char)SEGMENT_MAP[1] - 0x80); //display
digit on the 7-segment
                    }else{
                    spi_send((unsigned char)SEGMENT_MAP[0] - 0x80);
                }
                spi_send((unsigned char)0xF2); //select digit
                SHIFT_PORT |= LATCH; //toggle latch to copy data to storage register
                SHIFT_PORT &= ~LATCH;
                alternator = 0;
                }else{
                spi_send((unsigned char)SEGMENT_MAP[(recTicks2/24) % 10]);
                spi_send((unsigned char)0xF4);
```

```
                SHIFT_PORT |= LATCH;
                SHIFT_PORT &= ~LATCH;
                alternator = 1;
            }
        }
}
```

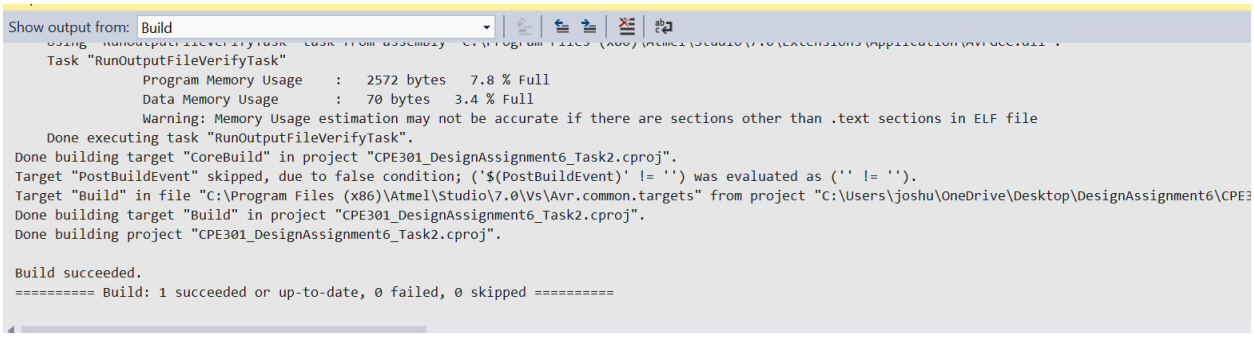
3. KICAD SCHEMATICS FOR TASK

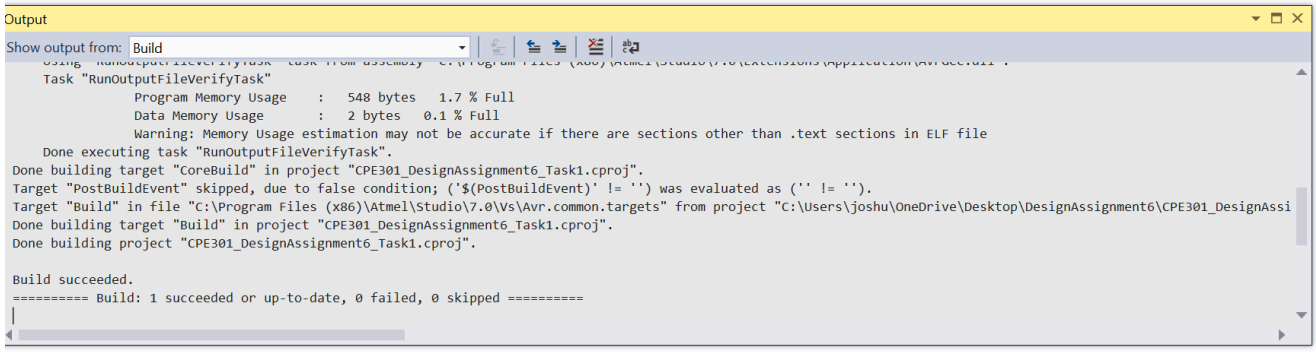




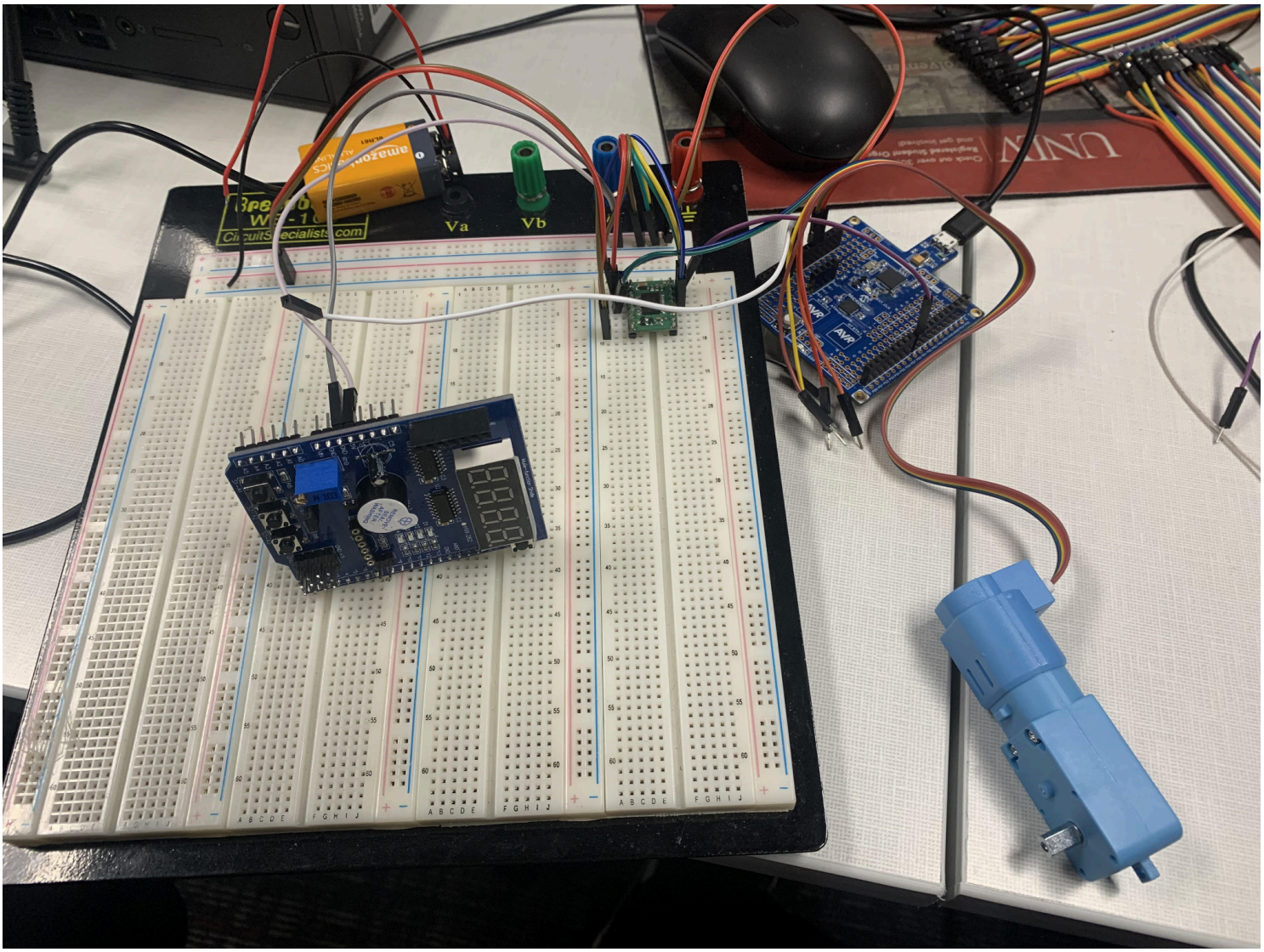
4. SCREENSHOTS OF EACH TASK OUTPUT

```
                Data Memory Usage       :   46 bytes   2.2 % Full
                Warning: Memory Usage estimation may not be accurate if there are sections other than .text sections in ELF file
    Done executing task "RunOutputFileVerifyTask".
Done building target "CoreBuild" in project "CPE301_DesignAssignment6_Task3.cproj".
Target "PostBuildEvent" skipped, due to false condition; ('$(PostBuildEvent)' != '') was evaluated as ('' != '').
Target "Build" in file "C:\Program Files (x86)\Atmel\Studio\7.0\Vs\Avr.common.targets" from project "C:\Users\joshu\OneDrive\Desktop\DesignAssig
Done building target "Build" in project "CPE301_DesignAssignment6_Task3.cproj".
Done building project "CPE301_DesignAssignment6_Task3.cproj".

Build succeeded.
========== Build: 1 succeeded or up-to-date, 0 failed, 0 skipped ==========
```





5. SCREENSHOTS OF EACH DEMO (BOARD SETUP)



6. VIDEO LINKS OF EACH DEMO

Task: https://www.youtube.com/playlist?list=PLZ09MA_uFt63me60r5XWtDUj2m21n4nUP

Student Academic Misconduct Policy
http://studentconduct.unlv.edu/misconduct/policy.html

"*This assignment submission is my own, original work*".
JOSHUA MARTINEZ